Engineering Product Specification

Rechargeable Lithium-Ion Polymer Pouch Cell (Model: 502535 / AP)

Standard: GB 31241-2022 **Batch / Date:** 2026.07.16 **Doc Rev:** 1.3 (Release)

1. SUPPLY CHAIN & IDENTIFICATION

Product Name: Rechargeable Lithium-ion Battery
Model Code: 502535 (Series: AP)
Producer: Dongguan AntPow New Energy Co., Ltd.
Manufacturer: Dongguan Aotai Digital Technology Co., Ltd.

2. RATING & OPERATIONAL WINDOW

Nominal Capacity: 400 mAh (0.2C discharge)
Nominal Voltage / Energy: 3.7 V / 1.48 Wh
Charge Cut-off Voltage: 4.20 V (±0.03 V)
Discharge Cut-off Voltage: 3.00 V (Typical recommended)

3. Detailed Electrical Characteristics

Parameter	Specification	Test Condition / Engineering Note
Rated Capacity	400 mAh (min. 380 mAh)	Standard CC/CV charge to 4.2 V, 0.2C discharge to 3.0 V @ 25 °C
Standard Charge	CC: 0.2C (80 mA) / CV: 4.2 V	Cut-off at charging current ≤ 0.02C (8 mA)
Maximum Fast Charge	1.0C (400 mA)	0.5C to 1.0C continuous; monitoring required above 0.5C
Max Discharge Current	1.0C (400 mA continuous)	Peak discharge up to 2.0C (800 mA) for ≤ 2 seconds
Internal Impedance	≤ 120 mΩ	AC 1 kHz test method after 50% SOC conditioning
Service Life Cycle	≥ 300 cycles	Residual capacity ≥ 80% under standard 0.2C cycling

4. Mechanical Dimensions & PCM Protection Subsystem

Dimension / Subsystem	Rating	Configuration & Thresholds
Physical Package	T: 5.0 mm × W: 25 mm × L: 35 mm	Industry 502535 pouch format; max initial thickness 5.2 mm
Integrated PCM Leadout	P+ (Red wire), P- (Black wire)	On-board PCB with top-mount B+/B- cell terminals and polyimide tape
Overcharge Protection	4.28 V ± 0.025 V	Automatic cutoff; recovery release threshold: 4.08 V ± 0.05 V
Overdischarge Cutoff	2.50 V ± 0.08 V	Hardware trip cutoff; recovery release threshold: 3.00 V ± 0.10 V
Operating Temperature	Charge: 0 ~ +45 °C / Discharge: -20 ~ +60 °C	Recommended long-term storage: +15 ~ +25 °C at 40–60% SOC

5. Safety & Handling Compliance (GB 31241-2022)

- **Prohibition:** Strictly avoid disassembly, crush, mechanical impact, puncture, or disposal in fire.
- **Swelling Caution:** Discontinue use immediately if the pouch exhibits observable expansion, severe swelling, or electrolyte odor.
- **Thermal & Moisture Hazard:** Do not place in ambient temperatures exceeding rated operating range. Forbidden to use after water immersion.